# Algebra 2 and Precalculus Final

Name: ______________________ Date: ______________________

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**Instructions:** This assessment has 50 questions spanning Algebra 2 and Precalculus. Show complete work for full credit. Unless stated otherwise, give exact answers (use radicals, fractions, and $\pi$ as needed). Clearly label graphs and justify reasoning for proofs but the rigor standards should not be extremely high, the main ideas should be correct and clear.

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1. Evaluate when $a = -5$ and $b = \dfrac{3}{4}$: $\quad (a-b)^2 - 2ab + 3a$. (8)

2. Simplify completely: $\quad \dfrac{x^2-9}{2x} \cdot \dfrac{4x^2}{x^2-x-6}$, with domain restrictions. (8)

3. For each set with usual addition and multiplication, state whether it forms a field. If not, name one property that fails. (10)

   (a) All nonzero rational numbers $\mathbb{Q} \setminus \{0\}$

   (b) The even integers $\{2k : k \in \mathbb{Z}\}$

   (c) Real numbers $\mathbb{R}$

4. The sum of 17 consecutive integers is 714. Find the median (middle) integer. (8)

5. A chemist wants 24 liters of a 35% acid solution by mixing a 20% solution with a 50% solution. How many liters of each should be used? (10)

6. Lines $\ell_1$ and $\ell_2$ share the same $y$-intercept and have slopes $m$ and $-3m$ (with $m \neq 0$). Find the sum of the $x$-intercepts of $\ell_1$ and $\ell_2$ in terms of $b$ and $m$. (8)

7. Find $k$ so that the line $kx + 4y = 7$ is perpendicular to $3x - 2y = 5$. Then find the $y$-intercept of $kx + 4y = 7$. (8)

8. Given $A(2,-3)$ and $B(-6,5)$, find the distance $AB$, the midpoint of $\overline{AB}$, and the equation of the line through $A$ and $B$ in standard form. (10)

9. Let $f(x) = 2x - 1$ and $g(x) = x^2 - 3x + 2$. Compute $f(g(-2))$ and $g(f(5))$. (8)

10. Consider the relation $S = \{(-2, 1), (0, 1), (2, 1), (2, -3)\}$. Is $S$ a function? State the domain and range. (8)

11. Solve each equation. (10)

(a) $y^2 - 9y = 0$

(b) $2x^2 + 5x - 3 = 0$

(c) $4t^2 = 196$

12. Solve the inequality and show the solution on a number line: $x^2 - 5x - 14 \leq 0.$ (10)

13. Factor completely: $x^5 - y^5$ and $x^6 - y^6.$ (10)

14. Solve by factoring or substitution: $u^4 - 13u^2 + 36 = 0.$ (10)

15. Find the greatest common factor of $54x^6y^3$, $81x^4y^2$, and $45x^3y$. (8)

16. A rectangle's length exceeds its width by 7 cm. If both dimensions are increased by 3 cm, the area becomes $170\,\text{cm}^2$. Find the original dimensions. (10)

17. Solve for $x$: $\dfrac{x+4}{x-2} = 2 + \dfrac{5}{x-3}$. (10)

18. Let $h(x) = \dfrac{2x-3}{x+4}$. Find the domain, range, vertical asymptote(s), horizontal or oblique asymptote, and intercepts. (12)

19. Find constants $A$ and $B$ so that $\dfrac{7x+8}{(x-1)(x+2)} = \dfrac{A}{x-1} + \dfrac{B}{x+2}$. (10)

20. Decompose $\dfrac{5x^2 - 3x + 7}{x(x+1)^2}$ into partial fractions of the form $\dfrac{A}{x} + \dfrac{B}{x+1} + \dfrac{C}{(x+1)^2}$. (12)

21. Simplify each expression, stating any restrictions. (10)

(a) $\dfrac{x^2 - 25}{x^2 + 7x + 10}$

(b) $\dfrac{x^3 - 8}{x^2 - 4}$

22. Find the oblique asymptote of $f(x) = \dfrac{3x^2 - 2x + 5}{x - 4}$ using long division. (10)

23. Rationalize the denominator and simplify: $\dfrac{\sqrt{3}+\sqrt{5}}{\sqrt{3}-\sqrt{5}}$. (10)

24. Evaluate the telescoping sum: $\displaystyle\sum_{k=3}^{100} \frac{1}{\sqrt{k}+\sqrt{k-1}}$. (10)

25. Solve for real $x$: $5 - 2\sqrt{3x+1} = x$. (10)

26. Simplify: $\sqrt[3]{54} - \sqrt[3]{16} + \sqrt[3]{2}$. (8)

27. Find all complex numbers $z = a + bi$ satisfying $\overline{z} = z^2$. (12)

28. Compute and express in $a + bi$ form: $\dfrac{1}{i-2} + \dfrac{3}{1+2i}$. (8)

29. Prove: If $r \in \mathbb{Q}$ with $r \neq 0$ and $t \notin \mathbb{Q}$, then $rt \notin \mathbb{Q}$. (10)

30. Suppose $x^2 + px + q = 0$ has roots $\alpha$ and $\beta$ (nonzero and not equal to $-1$). Find a quadratic equation whose roots are $\dfrac{1}{\alpha + 1}$ and $\dfrac{1}{\beta + 1}$. (10)

31. Find the vertex and axis of symmetry of $y = 2x^2 - 8x + 11$. State whether the graph opens upward or downward. (8)

32. Find the standard-form equation of the parabola with focus $(2, -1)$ and directrix $y = 3$. (10)

33. For which real values of $m$ does $mx^2 + 4x + 9 = 0$ have two distinct real roots? Justify using the discriminant. (10)

34. A quadratic $y = ax^2 + bx + c$ has vertex $(3, -4)$ and passes through $(1, 8)$. Find $a$, $b$, and $c$. (12)

35. Solve for $x$ (state domain first): $\log_5(2x-3)+\log_5(x-1)=2.$ (10)

36. Evaluate the number of digits in $N$ where $\log_2(\log_3(N))=4.$ (10)

37. Find the domain and range of $f(x) = \ln(x^2 - 6x + 10)$ and determine whether $f$ is one-to-one on its domain. Justify. (12)

38. Let $f(x) = \dfrac{1}{x+1}$. Compute $f(f(f(2)))$. (10)

39. A payment of \$5000 is due in 8 years. At an annual effective interest rate of 6%, what is its present value? If instead money is compounded monthly at a nominal annual rate of 6%, what is the present value? (12)

40. Give exact values: $\sin\left(\frac{7\pi}{6}\right)$, $\cos(-\frac{2\pi}{3})$. (8)

41. Without graphing, determine all solutions with $0 \le \theta < 2\pi$ of $\quad 2\sin\theta\cos\theta = \dfrac{\sqrt{3}}{2}$. (10)

42. Verify the identity $\quad \dfrac{1-\cos 2x}{\sin x} = 2\sin x \quad$ on the set where both sides are defined. (10)

43. Find the exact values of $\sin 75°$ and $\cos 75°$ using angle addition formulas. (12)

44. Solve for all $x$ in $[0, 2\pi)$: $\cos(2x) + 2\sin x - 1 = 0.$ (12)

45. Prove the Law of Cosines for a triangle with sides $a, b, c$ opposite angles $A, B, C$: $c^2 = a^2 + b^2 - 2ab\cos C$. (12)

46. Circle $\mathcal{C}$ has center $(4, -1)$ and passes through $(7, 3)$. Find its equation and radius. (10)

47. For the ellipse $\dfrac{(x+2)^2}{25} + \dfrac{(y-1)^2}{9} = 1$, find the center, vertices, co-vertices, foci, and lengths of the axes. (12)

48. A hyperbola centered at the origin has asymptotes $y = \pm\dfrac{5}{12}x$ and a vertex at $(0, 7)$. Find its standard equation. (12)

49. Convert the polar point $\left(-6, \frac{5\pi}{4}\right)$ to rectangular coordinates, and convert $(-3, 3\sqrt{3})$ to polar with $0 \leq \theta < 2\pi$. (10)

50. Consider the parametric curve $x = 3\cos t + 2$, $y = 4\sin t - 1$ for $0 \leq t \leq 2\pi$. Sketch one cycle, indicate direction, and eliminate $t$ to obtain a rectangular equation. (12)